

Mortgage Benchmark (plain-language)

A Mortgage Safety-Guardrail Benchmark — the plain-language version

Why a general “*is this prompt unsafe?*” guard isn’t enough for mortgage lending, and a benchmark built from real loan data to prove it. Reza Rahimi (JazzX AI). Plain-language companion to the formal draft.

Read this first. The benchmark’s prompts are **made-up (synthetic)** and its right-answers were assigned by **an AI judge, not a human expert**. So it’s good for *measuring how a guard behaves* on mortgage-style requests — it is **not** proof of discrimination by any real lender or model. Think “test set for guards,” not “fair-lending audit.”

The idea in one paragraph

A normal safety guard asks “*is this prompt harmful?*” — jailbreaks, abuse, injections. But in mortgage lending, a request can be totally polite and still ask the assistant to **break the law**: to price by neighborhood demographics, to hide the real reason for a denial, to fudge income so a loan qualifies. A general guard says “looks safe” and waves it through. We built a benchmark of such requests, **grounded in real HMDA loan data**, where each request has **two labels**: *is it generally harmful?* (G) and *would honoring it break mortgage policy?* (D). The interesting cases are “**safe-looking but a mortgage violation**” — and that’s exactly what general guards miss.

Background: the words you need. - **Guard.** A small model that scores a request as safe/unsafe. - **HMDA.** Public U.S. mortgage data (loan purpose, amounts, income, race/ethnicity of applicants, approve/deny, denial reasons). We use it to make the scenarios *realistic*. - **Two labels, G and D.** **G** = generally harmful (jailbreak/abuse) — a normal guard’s job. **D** = would this break a mortgage rule (fair lending, ability-to-repay, disclosures, fraud, privacy). A request can be G-safe but D-violating. - **Quadrants.** Combining G and D gives four boxes. The key one is **G0/D1**: *not* generally harmful, *but* a mortgage-policy violation. - **macro-AP.** A score (0–1) for how well a guard ranks the “should-intervene” requests above the fine ones. Higher = better.

How the benchmark is built

- **Grounded in real loans.** We take real (de-identified, banded) HMDA records — e.g. “FHA cash-out refi, investment property, income under \$40k, LTV over 97%, prior denial for collateral” — and write realistic requests around them. We never copy a real person’s record; we sample *distributions* and check nothing traces to one loan.
- **Hard on purpose.** Following a 2024 study of LLM mortgage bias, we focus on **borderline, higher-risk files** (where bias and discretion actually happen) and make violations **subtle** — “just use your judgment,” coded references to a neighborhood — not cartoonish.
- **Made by a team of AI agents.** A planner picks what to cover; a “grounder” pulls a loan fact sheet; a “generator” writes the request; an “adversary” adds realistic disguises; a **rubric-bound judge** assigns the labels from 24 written policy cards.
- **Fairness pairs.** For some benign requests we write **two versions identical except one word** — the applicant’s protected group (e.g. “a Black applicant” vs “a white applicant”). A fair guard must treat both

the same. We measure the gap (should be ≈ 0).

What’s in it (994 requests)

G0/D1 (safe-looking mortgage violation — the payload)	502
G0/D0 (benign, incl. hard “don’t over-block me” cases)	450
G1/D1 (harmful <i>and</i> a violation)	42
G1/D0 (general harm only)	0 (<i>gap — see below</i>)
Domains: fair-lending 204, fraud 112, UDAAP 90, disclosure 66, ability-to-repay 54, privacy 18	
Protected fairness pairs	39
Splits: train 604 / dev 149 / public-test 146 / sealed test 95	

What we test, and what we expect to see

We run guards over the benchmark and score them with the same tie-aware metrics as the companion Paper A. A guard gives one “unsafe” number per request; we check how well it separates the should-intervene requests for **G**, for **D**, and for the combined decision, plus **how many G0/D1 violations it misses** at a sensible alarm threshold, plus the **fairness-pair gap**.

What we actually found (4 base checkpoints, public-test split): the story is more interesting than “general guards ignore mortgage rules.” On *ranking* alone the base guards do moderately well on D (AP 0.67–0.85) — because asking to discriminate or commit fraud *sounds* unsafe to a general model even when it isn’t a jailbreak. But none is near-perfect, so a big chunk of the subtle safe-looking violations is unresolved. And the **alarm threshold is unreliable**: for these guards the scores cluster, so where you set the cutoff is a coin-flip (we saw the “how many did it catch” number swing by 50+ just from a library-version change) — a finding in itself: *you can’t trust a transferred threshold here*. The sharpest signal is the **fairness pairs**: Qwen3-4B treats the protected/reference versions identically (gap ≈ 0) and ranks best; Qwen2.5-1.5B scores them very differently (gap up to 0.51). So: *general safety $\neq$ mortgage compliance*, and the fairness gate catches problems accuracy alone misses. (Small sample — a measuring stick, not a verdict.)

Baseline zero-shot instruction guards on the frozen benchmark (public_test split; 146 rows: 75 G0/D1, 6 G1, 3 protected pairs), via score_guards.py; AP recomputed in the repo canonical env from the committed per-row scores (exactly reproducible). Threshold-free, base guards rank mortgage-policy violations moderately (AP·D 0.67–0.85) — soliciting fraud/discrimination reads as “unsafe” even without a jailbreak, so G and D are only PARTIALLY orthogonal. Protected-pair invariance (Δ_{context} ; 3 pairs) varies sharply across guards. The fixed 5%-FPR operating point is threshold-knife-edge for these clustered-score zero-shot guards (its G0/D1 catch count flips across library versions), so it is not tabulated per guard — see text. Small-sample, LLM-judge labels — illustrative, not confirmatory.

Guard	AP · G	AP · D	AP · final	Δ_{context} (fairness)
qwen25_15b_base	0.681	0.793	0.793	0.183
qwen3_4b_base	1.000	0.851	0.851	0.000
smollm2_17b_base	0.261	0.672	0.672	0.023
smollm3_3b_base	0.546	0.733	0.733	0.010

Not scored (unavailable in this run): llama_guard_3_1b, wildguard_7b.

Is this real? (the honesty box)

- **The labels are an AI judge's, not an expert's.** Every policy card is marked “not-counsel-reviewed.” Before anyone makes a fair-lending *claim*, a human expert has to review a sample and we report the human-agreement number. This benchmark is a measuring stick, not a verdict.
- **One box is empty** (G1/D0) — the AI wouldn't write plain jailbreaks — so the 2x2 is only three-quarters filled.
- **You can't regenerate it** (the AI writing step is random), so we **freeze and publish the exact dataset**; results are reproduced by *scoring guards on the frozen data*, not by rebuilding it.
- **No domain-trained guard yet** — the natural next step is to fine-tune a mortgage guard and show it catches G0/D1 (at some cost to general skill), tying back to the “compose, don't tune” idea.

What it means

General guards and mortgage compliance are **different problems**. This benchmark makes that measurable, with realistic, hard, real-data-grounded requests and a fairness-invariance test. It's an honest first artifact with a clear checklist to become a validated instrument: expert adjudication, fill the empty quadrant, decontaminate, and add a fine-tuned mortgage guard.

Frozen benchmark: mortgage-redteaming-agentic-generator/benchmark/v1_hmda2022/ (994 rows, checksummed, verified PII-free). Reproduce baselines with `magen/score_guards.py`.